Electrostatics

Michael Remp

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Coulomb's Law & Electric Field

Source charges q — responsible for creating electric field/force

Test charge Q — charge used to measure the exisiting electric field/force

Principal of Superposition — interaction between two charges is independant of other charges

Coulomb's Law

$$\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{qQ}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}} \tag{1}$$

Where $\vec{\boldsymbol{\mathscr{r}}} = \vec{r} - \vec{r'}$ is the **separation vector**, $\vec{r}$ & $\vec{r'}$ are **position vectors** for test charge and source charge respectively

Permitivity of free space $\epsilon_0 \equiv 8.85 \times 10^{-12}\frac{C^2}{N \cdot m^2}$

The Electric Field

Force due to multiple charges,

$$\vec{F} = \vec{F}_{total} = \vec{F}_1 + \vec{F}_2 = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1 Q}{\mathscr{r}_1^2}\hat{\boldsymbol{\mathscr{r}}}_1 + \frac{q_2 Q}{\mathscr{r}_2^2}\hat{\boldsymbol{\mathscr{r}}}_2 + \dots\right)$$
$$\vec{F} = \frac{Q}{4\pi\epsilon_0}\left[\frac{q_1}{\mathscr{r}_1^2}\hat{\boldsymbol{\mathscr{r}}}_1 + \frac{q_2}{\mathscr{r}_2^2}\hat{\boldsymbol{\mathscr{r}}}_2 + \dots\right]$$

$\vec{E} \equiv$ electric field of source charges.

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\left[\frac{q_1}{\mathscr{r}_1^2}\hat{\boldsymbol{\mathscr{r}}}_1 + \frac{q_2}{\mathscr{r}_2^2}\hat{\boldsymbol{\mathscr{r}}}_2 + \dots\right] = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{\mathscr{r}_i^2}\hat{\boldsymbol{\mathscr{r}}}_i \tag{2}$$

then,

$$\vec{F} = Q\vec{E}(\vec{r}) \tag{3}$$

$\vec{E}(\vec{r}) \equiv$ force per unit charge on a test charge at point P

Example 2.1 — Find the electric field a distance z above the midpoint between two equal charges q, a distance d apart.

$$\text{Point charges} — \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i^2}\hat{\boldsymbol{\mathscr{r}}}$$

Let $\vec{E}_1$ be the electric field of the left charge and $\vec{E}_2$ be the field of the right charge.

By symmetry, the horizontal components cancel, so find the vertical components and add.

$$E_{1z} : q = q, \quad \mathscr{r} = \sqrt{z^2 + (d/2)^2} \quad \vec{\boldsymbol{\mathscr{r}}} = (d/2)\hat{x} + z\hat{z} \quad \hat{\boldsymbol{\mathscr{r}}} = \frac{\vec{\boldsymbol{\mathscr{r}}}}{\mathscr{r}} = \frac{z}{\sqrt{z^2 + (d/2)^2}}\hat{z}$$

$$\vec{E}_1 z = \frac{1}{4\pi\epsilon_0}\frac{q}{\sqrt{z^2 + (d/2)^2}^2}\frac{z}{\sqrt{z^2 + (d/2)^2}}\hat{z} = \frac{1}{4\pi\epsilon_0}\frac{qz}{(z^2 + (d/2))^{3/2}}\hat{z}$$

$$\vec{E_{1z}} = \vec{E_{2z}} \longrightarrow \vec{E_{1z}} + \vec{E_{2z}} = \boxed{\frac{1}{4\pi\epsilon_0}\frac{2qz}{(z^2 + (d/2))^{3/2}}\hat{z}}$$

Continuous Charge Distributions

If the charge is continuously distributed over some region, the sum in (Eq. 2) becomes an integral.

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{1}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}}dq \tag{4}$$

Electric field of a **line charge**, $dq \to \lambda dl'$ where dl' is an element of length along the line,

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\lambda(\vec{r})}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}}dl' \tag{5}$$

Electric field for a **surface charge**, $dq \to \sigma da'$ where da' is an element of area on the surface,

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\sigma(\vec{r})}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}}da' \tag{6}$$

Electric field for a **volume charge**, $dq \to \rho dl'$ where $d\tau'$ is an element of volume,

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\vec{r})}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}}d\tau' \tag{7}$$

Example 2.2 — Find the electric field a distance z above the midpoint of a straigh line segment of length $2L$ that carries a uniform line charge λ

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\lambda(\vec{r'})}{\mathscr{r}^2}\hat{\boldsymbol{\mathscr{r}}}dl'$$

Horizontal components are going to cancel

$dq = \lambda dl'$, dl' is dx along x-axis $\quad \vec{r} = z\hat{z}$ (test position vector) $\quad \vec{r'} = x\hat{x}$ (source position vector)

$$\vec{\boldsymbol{\mathscr{r}}} = z\hat{z} - x\hat{x} \quad \mathscr{r} = \sqrt{z^2+x^2} \quad \hat{\boldsymbol{\mathscr{r}}} = \frac{z\hat{z}-x\hat{x}}{\sqrt{z^2+x^2}}$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int_{-L}^{L}\frac{\lambda}{\left(\sqrt{z^2+x^2}\right)^2}\frac{z\hat{z}-x\hat{x}}{\sqrt{z^2+x^2}}dx$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\lambda\left[z\hat{z}\int_{-L}^{L}\frac{1}{\left(z^2+x^2\right)^{3/2}}dx - \hat{x}\int_{-L}^{L}\frac{x}{\left(z^2+x^2\right)^{3/2}}\right]$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\lambda\left[z\hat{z}\left(\frac{x}{z^2\sqrt{z^2+x^2}}\right)\Bigg|_{-L}^{L} - \hat{x}\left(-\frac{1}{\sqrt{z^2+x^2}}\right)\Bigg|_{-L}^{L}\right]$$

$$\boxed{\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\frac{2L\lambda}{z\sqrt{z^2+L^2}}\hat{z}}$$

Problem 2.6 — Find the electric field a distance z above the center of a flat circular disk of radius R that carries a uniform surface charge σ

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}}dq \qquad dq = \sigma dA \qquad dA = rdr\cdot d\phi \qquad dq = \sigma r dr d\phi$$

each ring: $\vec{\boldsymbol{\mathscr{r}}} = \vec{r} - \vec{r'} = (z\hat{z} + 0\hat{r}) - (0\hat{z} + r\hat{r}) = z\hat{z} - r\hat{r} \qquad r = \sqrt{z^2+r^2} \qquad \hat{\boldsymbol{\mathscr{r}}} = \dfrac{z\hat{z}-r\hat{r}}{\sqrt{z^2+r^2}}$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int_{\phi=0}^{2\pi}\int_{r=0}^{R}\frac{1}{\left(\sqrt{z^2+r^2}\right)^2}\frac{z\hat{z}-r\hat{r}}{\sqrt{z^2+r^2}}\sigma r dr d\phi \qquad \hat{r}\to 0$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int_{\phi=0}^{2\pi}\int_{r=0}^{R}\frac{z}{\left(z^2+r^2\right)^{3/2}}\hat{z}\sigma r dr d\phi$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\sigma z 2\pi\int_{r=0}^{R}\frac{r}{\left(z^2+r^2\right)^{3/2}}\hat{z}dr$$

$$\boxed{\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\sigma z 2\pi\left[\frac{1}{z} - \frac{1}{\sqrt{z^2+R^2}}\right]\hat{z}}$$

Divergence & Curl of Electrostatic Fields

The flux of $\vec{E}$ through surface S — $\Phi_E \equiv \int_s \vec{E} \cdot d\vec{a}$

The charge enclosed in a closed surface — Q_{enc}

Gauss's Law

The flux through a spherical surface of radius r,

$$\oint \vec{E} \cdot d\vec{a} = \int \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \cdot (r^2 \sin\theta dr d\theta d\phi \hat{r}) = \frac{1}{\epsilon_0} Q_{enc} \tag{8}$$

For any closed surface,

$$\boxed{\oint \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}} \quad \textbf{Gauss's Law in integral form} \tag{9}$$

Apply divergence theorem to find differential form,

$$\oint \vec{E} \cdot \vec{a} = \int_V (\vec{\nabla} \cdot \vec{E}) d\tau$$

Rewrite Q_{enc} in terms of charge density ρ,

$$Q_{enc} = \int_V \rho d\tau$$

So,

$$\int_V (\vec{\nabla} \cdot \vec{E}) d\tau = \frac{Q_{enc}}{\epsilon_0} = \int_V \frac{\rho}{\epsilon_0} d\tau$$

Since this is true for any volume, we can drop the integrands.

$$\boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}} \quad \textbf{Guass's Law in differential form} \tag{10}$$